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Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and the Bayesian evidence in early-phase clinical trials

Puneet Velidi[1], Zhengxiao Wei[1], Evan Strasdin, Shreena Nisha Kalaria, Yimeng Liu, Céline M. Laumont, Brad H. Nelson and Farouk S. Nathoo

Abstract.

Concerns about the misuse and misinterpretation of p-values and statistical significance have motivated alternatives for quantifying evidence. We define a generalized form of Jeffreys's approximate objective Bayes factor (*eJAB*), a one-line calculation that is a function of the p-value, sample size, and dimension of the test. We establish conditions under which *eJAB* is model-selection consistent and verify them for common statistical tests as well as through empirical analysis of several replication studies. We assess finite-sample approximation accuracy by comparing *eJAB* with Markov chain Monte Carlo computed Bayes factors. Simulation studies compare the decision error probabilities of *eJAB* with test-statistic Bayes factors (TSBF; Johnson, 2005; Johnson, Pramanik and Shudde, 2023), BIC approximations, and classical NHST. We also discuss a modification of *eJAB* for one-sided tests and establish its model selection consistency. An exploratory reanalysis of 16,563 findings from 1,790 early-phase ClinicalTrials.gov trials flags 493 Bayes/NHST contradictions. A stronger eJAB > 3 screen yields 11 numerical reports and nine censored bounds; we examine their hypothesis contexts to identify illustrative Jeffreys–Lindley disagreements.

Key words and phrases: Jeffreys's approximate objective Bayes factor, Jeffreys–Lindley paradox, null-hypothesis significance testing, p-values, replicability.

1. INTRODUCTION

Concerns about the replicability of scientific findings, arising in part from the overreliance on null-hypothesis significance testing (NHST) at conventional p-value thresholds (Ioannidis, 2005), have prompted calls for more stringent standards of evidence (Benjamin et al., 2018; Johnson, 2013), more judicious interpretation of statistical significance (Wasserstein and Lazar, 2016),

Puneet Velidi: MSc Student, Department of Mathematics and Statistics, University of Victoria (e-mail: pvelidi@uvic.ca). Zhengxiao Wei: PhD Student, Department of Mathematics and Statistics, University of Victoria (e-mail: zhengxiao@uvic.ca). Evan Strasdin: MSc Student, Department of Mathematics and Statistics, University of Victoria (e-mail: evanastrasdin@uvic.ca). Shreena Nisha Kalaria: MSc Student, Deeley Research Centre, British Columbia Cancer Agency (e-mail: skalaria@bccrc.ca). Yimeng Liu: MSc Student, Department of Mathematics and Statistics, University of Victoria (e-mail: yimengliu@uvic.ca). Céline M. Laumont: Staff Scientist, Deeley Research Centre, British Columbia Cancer Agency (e-mail: claumont@bccrc.ca). Brad H. Nelson: Scientific Director, Deeley Research Centre, British Columbia Cancer Agency (e-mail: bnelson@bccrc.ca). Farouk S. Nathoo: Professor, Department of Mathematics and Statistics, University of Victoria (e-mail: nathoo@uvic.ca), Corresponding Author.

[1]Puneet Velidi and Zhengxiao Wei contributed equally to this work.

and alternative measures for quantifying statistical evidence (Berger and Sellke, 1987; Evans, 2016; Grünwald, 2024; Halsey, 2019). One such measure is the Bayes factor—the ratio of marginal likelihoods under competing hypotheses (Kass and Raftery, 1995; Kass and Wasserman, 1995; Ly, Verhagen and Wagenmakers, 2016; Morey, Romeijn and Rouder, 2016; Raftery, 1995; Wagenmakers, 2007). The evidential behavior of p-values and Bayes factors has been studied through stochastic ordering in sample size and effect size, with implications for the Jeffreys–Lindley paradox, under which the two approaches can yield conflicting conclusions (Held and Ott, 2016, 2018; Huisman, 2023; Wagenmakers and Ly, 2023; Wei, Nathoo and Masson, 2023). An important practical consideration, however, is the computational complexity of Bayes factors. since they require integration over prior distributions, often via numerical methods such as Markov chain Monte Carlo (MCMC) with model-specific tuning (JASP, 2026; Morey and Rouder, 2026; Sinharay and Stern, 2005). Approximate Bayes factors offer a computationally convenient alternative when exact or full marginal-likelihood calculations are difficult. Such approximations can be derived from asymptotic theory, information criteria, p-values and test statistics, or summary statistics.

In an early but underappreciated contribution, Jeffreys (1936) developed an approximate Bayes factor, referred to here as JAB, for testing a point-null hypothesis against a composite alternative under a normal unit-information prior centered at the maximum likelihood estimate (MLE). The resulting expression contains the Wald statistic in the exponential term and the effective sample size in a square-root term. Through a detailed reexamination of Jeffreys's work, Wagenmakers (2022) further simplified the exponential component using a piecewise approximation to the quantile function associated with the Wald statistic in JAB. The finite-sample form depends on the prior centering; for the one-sided construction below, we use a symmetric local prior centered at the null value. A one-sided version of JAB can then be obtained by multiplying the two-sided JAB by a factor reflecting the posterior mass assigned to the theoretically predicted direction (Marsman and Wagenmakers, 2017). More recently, Wagenmakers, Bartoš and Aust (2026) showed that, for single-parameter Wald-type tests, certain forms of JAB can be interpreted as approximate e-processes. Because e-processes support sequential monitoring and optional stopping while controlling the probability of accumulating misleading evidence against the null hypothesis, this connection suggests that JAB may also possess useful approximate frequentist error-control properties in sequential settings.

Another influential approximation is based on the Bayesian information criterion (BIC). For single-parameter tests, the Bayes factor approximation is asymptotically equivalent to JAB, with the likelihood-ratio test statistic replacing the Wald statistic. A related refinement is proposed by Bickel (2025), which incorporates a finite sample-size correction term into the BIC by raising the likelihood ratio to a correction-dependent power, thereby reducing evidential exaggeration under the null hypothesis.

An alternative line of research develops test-statistic Bayes factors (TSBFs) in closed form from the sampling distributions of test statistics under local alternatives, rather than treating them as approximations to full data Bayes factors (Johnson, 2005, 2008; Yuan and Johnson, 2008). More recently, the Bayes factor function has retained this core structure as a family of TSBFs evaluated over alternative hypotheses indexed by a standardized effect size, while extending the class of nonlocal priors used to define these alternatives (Datta et al., 2025a,b; Johnson, Pramanik and Shudde, 2023). For each fixed nonzero effect size, the resulting Bayes factor function is pointwise model-selection consistent.

Building on these developments, we derive an extended version of JAB, hereafter *eJAB*, using the Savage–Dickey density ratio representation under a generalized unit-information prior. This extension incorporates parameter dimension, accounts for finite-sample corrections, and facilitates applications to nonparametric testing. Our work differs from Bickel (2025) in its consideration of p-values beyond those from the likelihood ratio test and in our explicit statement of conditions for model-selection consistency. For the latter, Bickel (2025) demonstrates asymptotic equivalence to the BIC so that model-selection consistency arises under the conditions specified in Neath and Cavanaugh (2012). Our work also differs from Bartoš and Wagenmakers (2023), Rostgaard (2023), and Wagenmakers (2022) primarily in the formulation of the generalization and the development of its theoretical justification. We assess the finite-sample performance of *eJAB* across ten common parametric and nonparametric tests. The results suggest that *eJAB* is a broadly applicable and reliable approach to approximate the Bayes factor. We then apply *eJAB* to 16,563 reported primary- and secondary-outcome findings from 1,790 early phase I, phase I, and phase II trials registered on https://ClinicalTrials.gov (CTG), accessed on August 7th, 2025, and examine the profile of contradictions where NHST ($p \leq \alpha$) and $eJAB_{01} > 1$ (favoring the null) in recent clinical trials. We flag 493 such results and report the distribution of statistical evidence separately across the three early-phase classifications for

primary and secondary outcomes in CTG. A stronger eJAB > 3 screen yields 20 records, whose numerical reports, censored bounds, and hypothesis contexts are distinguished in the clinical analysis.

2. METHODOLOGY

DEFINITION 2.1. To test the hypotheses $\mathcal{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ versus $\mathcal{H}_1 : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$, we define *eJAB* as

$$eJAB_{01} = \sqrt{n} \exp\left\{ -\frac{1}{2} \frac{n^{1/q} - 1}{n^{1/q}} Q_{\chi^2_q}(1-p) \right\},$$

where q is the dimension of the test, n is the sample size, $Q_{\chi^2_q}(\cdot)$ is the quantile function of the chi-squared distribution with q degrees of freedom (df), and p is the p-value from an NHST. We further define $eJAB_{10} = 1 \,/\, eJAB_{01}$.

DEFINITION 2.2. For a scalar right-sided test of $\mathcal{H}_0 : \theta = \theta_0$ versus $\mathcal{H}_+ : \theta > \theta_0$, let p_1 denote the one-sided p-value and define $p = 2\min\{p_1, 1-p_1\}$. The corresponding *eJAB* is given by

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1-p_1)} \exp\left\{ -\frac{1}{2} Q_{\chi^2_1}(1-p) \right\}.$$

The left-sided test is handled analogously.

THEOREM 2.3. *Assuming the NHST p-value satisfies two regularity conditions,*

- ***R1***: $p \xrightarrow{D} \mathrm{Unif}(0,1)$ *as* $n \to \infty$ *under* $\mathcal{H}_0$,
- ***R2***: $D_n = -\sqrt{n} \cdot p \ln p \xrightarrow{P} 0$ *as* $n \to \infty$ *under* $\mathcal{H}_1$,

then,

$$\operatorname*{plim}_{n\to\infty} eJAB_{01} = \begin{cases} \infty & \text{under } \mathcal{H}_0, \\ 0 & \text{under } \mathcal{H}_1. \end{cases}$$

The proof is provided in Appendix D.

When $q = 1$, the two *eJAB* definitions are related by

$$eJAB_{0+} = \frac{n^{\frac{1}{2(1-n)}}}{2(1-p_1)} eJAB_{01}^{\frac{n}{n-1}}.$$

Under $\mathcal{H}_0$, **R1** implies $eJAB_{01} \xrightarrow{P} \infty$. Since $n^{\frac{1}{2(1-n)}} \to 1$ and $(2(1-p_1))^{-1} \geq 1/2$, we have $eJAB_{0+} \xrightarrow{P} \infty$. Under $\mathcal{H}_+$, **R2** implies $eJAB_{01} \xrightarrow{P} 0$. Assuming additionally **R3**: $(1-p_1)^{-1} = \mathcal{O}_p(1)$ under $\mathcal{H}_+$, it follows that $eJAB_{0+} \xrightarrow{P} 0$. Therefore, the one-sided *eJAB* is also model-selection consistent.

FACT. We can derive *eJAB* as an approximation using a generalization of the multivariate normal unit-information prior for the parameters under test,

$$\boldsymbol{\theta} \sim \boldsymbol{\mathcal{N}}_q\left(\hat{\boldsymbol{\theta}},\ n^{1/q} \cdot \mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})\right),$$

where $\hat{\boldsymbol{\theta}}$ is the MLE, and $\mathbf{J}_n(\hat{\boldsymbol{\theta}})$ is the observed information matrix of size $q \times q$. Under an asymptotic Gaussian approximation, the posterior density takes the form, $\pi(\boldsymbol{\theta} \mid \boldsymbol{y}, \mathcal{H}_1) \approx$

$$(2\pi)^{-\frac{q}{2}} |\mathbf{J}_n(\hat{\boldsymbol{\theta}})|^{\frac{1}{2}} \exp\left\{ -\frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^\top \cdot \mathbf{J}_n(\hat{\boldsymbol{\theta}}) \cdot (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \right\}.$$

Assuming that the prior density for any nuisance parameters under $\mathcal{H}_0$ is everywhere equal to the conditional prior density for those parameters under $\mathcal{H}_1$, given the null value of the parameter under test, the Bayes factor (BF) can be expressed as the Savage–Dickey density ratio (Mulder, Wagenmakers and Marsman, 2022),

$$BF_{01} = \frac{\pi(\boldsymbol{\theta}_0 \mid \boldsymbol{y}, \mathcal{H}_1)}{\pi(\boldsymbol{\theta}_0 \mid \mathcal{H}_1)} \approx \sqrt{n} \exp\left\{ -\frac{1}{2} \frac{n^{1/q} - 1}{n^{1/q}} W \right\},$$

where $\boldsymbol{y}$ represents the data, and W is the Wald statistic. When the p-value is obtained from a Wald test, substituting $W = Q_{\chi^2_q}(1-p)$ yields $eJAB_{01}$ in Definition 2.1.

The prior underlying the approximate BF interpretation is centered around the MLE $\hat{\boldsymbol{\theta}}$ with $|\mathrm{Var}(\boldsymbol{\theta})| = n\,|\mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})|$. When $q = 1$, this is the unit-information prior. More generally, the volume of prior probability ellipsoids around the MLE is $\mathcal{O}(n^{(1-q)/2})$. The prior distribution is asymptotically diffuse relative to the likelihood.

Setting $a_n = n^{\frac{1}{q}}$ yields $\sqrt{n}$ growth in the evidence for $\mathcal{H}_0$, so that under eJAB the rate of evidence accumulation with sample size towards $\mathcal{H}_0$ is independent of the dimension q. When $q > 1$ (F- and chi-squared tests) and effect sizes are small, this avoids the overly conservative behavior of the BIC approximation to the Bayes factor in favor of the null hypothesis (as detailed in the `R` Markdown files in the GitHub Simulations directory), while retaining model-selection consistency under R1 and R2.

FACT. When the p-value is obtained from a likelihood ratio test and $q = 1$, *eJAB* reduces to the evidential BIC method of Bickel (2025). This approximation is derived by applying a finite-sample correction to the BIC approximation to the BF, which amounts to raising the likelihood ratio to the power $(n-1)/n$. Model-selection consistency for the evidential BIC follows from its asymptotic equivalence to BIC together with standard consistency results for BIC (Bickel, 2025; Neath and Cavanaugh, 2012). By contrast, the conditions of Theorem 2.3 are broadly applicable: they cover nonparametric tests, provided **R1** and **R2** hold.

FACT. Applying the directional correction of Wagenmakers (2022, pp. 20–22) yields Definition 2.2. Unlike the MLE-centered prior used in the two-sided

eJAB (Definition 2.1, $q = 1$), we use a symmetric local prior centered at θ_0, thereby eliminating the $(n-1)/n$ factor in the exponent of $eJAB_{0+}$. The directional correction uses a prior probability of 1/2 and an approximate posterior probability of $1 - p_1$ in the direction of the alternative, assuming a relatively vague local prior and moderate sample sizes. Appendix C.5 provides implementation details for the clinical-trial analysis.

3. SIMULATION STUDIES

We conduct 12 simulation studies across ten parametric and nonparametric tests to evaluate the regularity conditions and finite-sample accuracy. For each design, 5,000 datasets are generated under the null hypothesis and the alternative hypothesis with small, medium, and large effect sizes across various sample sizes. The designs include:

1. Two-sample t-test;
2. Simple linear regression;
3. Simple logistic regression;
4. One-way analysis of variance (ANOVA);
5. One-way repeated-measures ANOVA (rANOVA) with high and low intraclass correlations $\rho \in \{.9, .2\}$;
6. Chi-squared tests for independence under multinomial (total count fixed) and product-multinomial (row counts fixed) designs;
7. Cox proportional-hazards regression with right censoring;
8. Wilcoxon signed-rank test;
9. Mann–Whitney U-test;
10. Kruskal–Wallis H-test.

For nonparametric tests (items 8 through 10), we simulated using right-skewed and heavy-tailed (non-Gaussian) distributions. Detailed simulation settings, additional designs (e.g., two-way ANOVA and large contingency tables), and the `R` code for the subsequent subsections are available in the GitHub repository (see files within 'Theorem', 'Simulations' [parametric], and 'Supplementary Information' [nonparametric]).

We used p-values from the t-test for items 1 and 2; the F-test for items 4 and 5; the Wald test for items 3 and 7; and the asymptotic null distribution for item 6. The p-values for 8–10 are based on asymptotic distributions. The effective sample size n was taken as the number of uncensored observations for survival analysis (Volinsky and Raftery, 2000) and as the number of independent observations (i.e., the product of the number of participants and one less than the number of conditions) for rANOVA (Masson, 2011; Nathoo and Masson, 2016). For the other designs, n always refers to the total number of observations or counts. Among the tests with multiple *df* ($q > 1$) are items 4 through 6, and 10.

3.1 Verifying the Regularity Conditions

Condition **R1** of Theorem 2.3 is satisfied whenever the p-value is computed from an exact or asymptotic null distribution with a continuous cumulative distribution function.

It is not difficult to verify that **R2** is insufficient to ensure model-selection consistency of the BIC approximation to the BF,

$$BF_{01}^{(\mathrm{BIC})} = n^{\frac{q}{2}} \exp\left\{-\frac{1}{2} Q_{\chi^2_q}(1-p)\right\}.$$

Using the asymptotic expansion of the incomplete gamma function (e.g., Daouia, Stupfler and Usseglio-Carleve, 2025, Lemma A.2), $\frac{1}{2} Q_{\chi^2_q}(1-p) = -\ln p + (\frac{q}{2} - 1)\ln(-\ln p) - \ln\Gamma(\frac{q}{2}) + o(1)$. It follows that $BF_{01}^{(\mathrm{BIC})} \approx \Gamma(\frac{q}{2})\, n^{\frac{q}{2}} \cdot p\,(-\ln p)^{1-\frac{q}{2}}$, which requires a substantially stronger condition than **R2** for model-selection consistency when $q > 1$.

We also examine **R2** empirically, using realized values of D_n from four large-scale replication projects for results whose original study and every replication attempt reached $p \le 0.005$, for which we assume $\mathcal{H}_1$ is true. Appendix E of the Supplementary Material describes the study; based on the realized value at the largest sample size, **R2** seems feasible.

3.2 Accuracy of the Bayes-Factor Approximation

To assess the finite-sample accuracy of an approximation to the BF, we compared *eJAB* with alternative computations in seven studies based on the parametric designs, using fixed representative sample sizes. In this subsection, item 7 represents a parametric survival regression model with an exponential baseline and expected censoring fractions of 23% under the null, and 21%, 19%, and 18% under increasing effect sizes. The MCMC-based BFs were computed by sampling the marginal posterior density using '*rstan*' (Stan, 2025), applying logspline density estimation (Kooperberg and Stone, 1992), and calculating the Savage–Dickey density ratio. In these cases, the model underlying the MCMC sampler assumed the same prior as *eJAB*, so discrepancies reflect errors of approximation rather than differences in prior.

Pairs of $(\ln eJAB_{10},\ \ln BF_{10})$ are color- and shape-coded by the p-value. Overall, *eJAB* tracks the MCMC-based BFs closely in Figure 1. Some inaccuracy, $\ln eJAB_{10} < \ln BF_{10}$, appears for logistic regression when evidence favors $\mathcal{H}_1$. Replacing *eJAB* with the BIC approximation to the BF performs well when $q = 1$, but marked severe discrepancies for $q > 1$, indicating that *eJAB* better approximates BF for ANOVA and rANOVA.

In sum, *eJAB* provides an accurate approximation to Bayes factors across a range of designs and effect sizes for parametric tests.

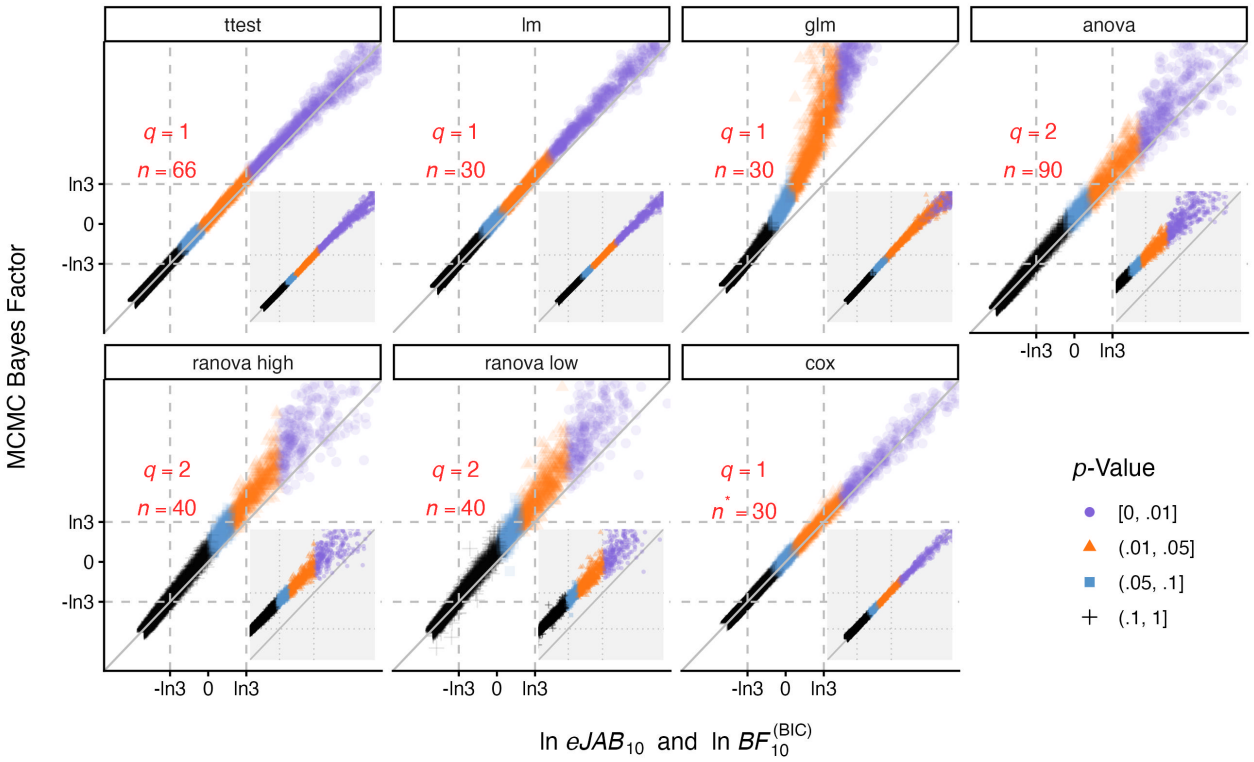


FIG 1. Comparison of the natural logarithms of $eJAB_{10}$ (main plots) and the BIC approximation (gray-background insets) with the MCMC-based Bayes factor. Red text indicates the test dimension q and effective sample size n, except for the Cox model, for which n^* denotes the number of observations rather than the number of events. Colors and shapes represent descriptive p-value intervals; NHST decisions use $p \leq \alpha$. Dashed lines indicate Bayes factors of 1/3 and 3.

3.3 Bayesian Hypothesis Tests Using Nonparametric Statistics

Classical nonparametric tests do not specify a sampling distribution for the data, so marginal likelihoods are not defined. As such, the Bayes factor can be defined based on the marginal likelihood of the test statistic rather than on the data (Yuan and Johnson, 2008). We compared *eJAB* with Bayes factors constructed from nonparametric statistics (*BFNP*) across items 8 through 10. Appendix B reports sequences of box plots depicting how the sampling distributions of *eJAB* and *BFNP* vary with effect size and sample size. Figures depicting these results, a detailed description of the experiments, and the associated `R` Markdown files are available in the GitHub repository (see the 'Supplementary Information' directory).

Under $\mathcal{H}_1$, when effects and samples are both relatively small, *BFNP* tends to provide stronger evidence for $\mathcal{H}_1$ than *eJAB*. With larger effects or larger samples, the two methods produce comparable results. Under $\mathcal{H}_0$, *eJAB* effectively quantifies evidence for $\mathcal{H}_0$.

3.4 Equally Weighted Decision Error Profiles

For each effect size θ and sample size n, we compare a simple null (effect zero) with a simple alternative (effect θ). Under 0–1 loss, the equally weighted decision error (EWDE) for decision rule d is

$$\text{EWDE}_d(\theta, n) = \frac{\alpha_d(\theta, n) + \beta_d(\theta, n)}{2},$$

where α_d is the type I error probability under the null and β_d is the type II error probability at the specified alternative, equal to one minus power. Define $\Delta E = \text{EWDE}_{\text{comparison}} - \text{EWDE}_{\text{eJAB}}$. The profiles display the normalized EWDE comparison,

$$\text{EWDE}_{\text{norm}}(\theta, n) = \frac{\Delta E(\theta, n)}{M},$$

where $M = \max_{(\theta,n)\in\mathcal{G}} |\Delta E(\theta, n)|$ and $\mathcal{G}$ is the displayed grid for each panel. Positive values favor eJAB; negative values favor the comparison rule. Figure 2 compares *eJAB* with the $p \leq .005$ rule (A), the BIC Bayes-factor approximation (B), and true- and estimated-effect TSBF rules (C–D). Panel B uses $BF_{01}^{\text{BIC}} = n^{q/2}\exp(-\Lambda/2)$, where Λ is the fitted likelihood-ratio statistic, across the six parametric-likelihood tests with available BIC comparisons. Cox regression and the three rank tests are excluded from this panel. Panel C uses the true standardized effect ω with $\tau_\omega^2 = n\omega^2/q$. For Panel D, $\hat{\omega} = \sqrt{X^2/n}$ and $\hat{\tau}^2 = X^2/q$, where X^2 is the observed chi-squared statistic. The standalone

$p \leq .05$ comparison is provided in Supplementary Figure S3 (Appendix B).

For Pearson chi-squared tests of independence in 2×2 and 3×3 tables, *eJAB* generally has lower EWDE than estimated-ω TSBF outside small-sample or weak-signal regimes. Small 3×3 samples can favor TSBF even at larger effects. The broad pattern reverses for 5×5 tables: estimated-effect TSBF generally has lower EWDE, with *eJAB* gains concentrated in a band of weaker effects. With true ω, *eJAB* generally matches or underperforms TSBF; differences can be appreciable near weak signals. BIC comparisons show small absolute differences for the single-parameter tests. For ANOVA, repeated-measures ANOVA and chi-squared independence tests, eJAB's largest gains occur in weaker-effect bands that move toward zero as sample size increases. At stronger effects, BIC can have lower EWDE because both rules have high power and BIC has lower type I error; the differences diminish at larger samples. Against the NHST significance rule $p \leq .005$, *eJAB* gains in bands of weaker effects that move toward zero as sample size increases. NHST can have lower EWDE at stronger effects in smaller samples, particularly for multiparameter tests. At larger sample sizes, differences at stronger effects are small, while weak-signal bands can favor NHST.

Dashed curves of the form $\theta = c/\sqrt{n}$ hold $\sqrt{n}\theta$ constant. They preserve the alternative distribution when it depends on n and θ only through that quantity, but need not hold normalized EWDE constant because the *eJAB* threshold also depends on n. eJAB, BIC, and TSBF reject $\mathcal{H}_0$ for $BF_{01} < 1$ (not ties); NHST uses $p \leq \alpha$. Under fixed priors, equal model probabilities and 0–1 loss, threshold 1 minimizes prior-predictive error, but need not minimize EWDE. The estimated-effect TSBF is a plug-in sensitivity analysis.

4. THE JEFFREYS–LINDLEY PARADOX AND BAYES/NHST CONTRADICTIONS IN EARLY-PHASE CLINICAL TRIALS

We apply *eJAB* to ClinicalTrials.gov (CTG) results to illustrate the Jeffreys–Lindley paradox: a conventionally significant result can coexist with evidence favoring a point null under the specified Bayes-factor approximation. We retain the broader analysis of Bayes/NHST contradictions, defined by statistical significance and applicable eJAB > 1, and examine the stronger disagreements with eJAB > 3 in detail. Neither classification establishes that a null hypothesis is true or identifies a realized type I error.

Our objective is to characterize the overall strength of evidence in 16,563 primary- and secondary-outcome CTG findings from 1,790 early-phase trials and to identify evidential disagreements warranting closer examination. These results span 36 MeSH categories (e.g., neoplasms, cardiovascular diseases, and autoimmune diseases) and encompass eight statistical tests: one-sample t-test, two-sample t-test, linear regression, logistic regression, ANOVA, Wilcoxon signed-rank test, Mann–Whitney test, and Kruskal–Wallis test.

For each, we extracted the p-value, sample size, and parameter dimension, and computed the applicable eJAB. For the 427 explicitly one-sided scalar analyses, we used the null-centered one-sided *eJAB* in Definition 2.2, with $p = 2\min\{p_1, 1 - p_1\}$ and no $(n-1)/n$ factor in its exponent; for the remaining 16,136 two-sided/unrestricted analyses, we used $eJAB_{01}$. Statistical significance was defined as $p_1 \leq .025$ and $p \leq .05$, respectively, retaining reported inequalities as described in Appendix C.5. Throughout the clinical comparisons and plots, we use $p_{\text{NHST}} = p$ for two-sided/unrestricted tests and $p_{\text{NHST}} = 2p_1$ for one-sided tests. Thus $p_{\text{NHST}} \leq \alpha$ means $p_1 \leq \alpha/2$ for a one-sided test, preserving its specified direction. This comparison value can exceed one; the folded value $2\min\{p_1, 1-p_1\}$ is used only to reconstruct the unsigned statistic inside eJAB. We excluded survival analyses and chi-squared tests because CTG usually omits the number of uncensored observations and full contingency-table dimensions. Preprocessing to address misreported and left-censored p-values, sample sizes, and test labels (Miron, Gonçalves and Musen, 2020) is described in Appendix A.

Within phase g, let N_g count all eligible analyses, $S_g(\alpha)$ those with $p_{\text{NHST}} \leq \alpha$, and $C_g(\alpha)$ those also having applicable eJAB > 1. Figure 3A plots the overall proportion $C_g(\alpha)/N_g$; Figure 3C plots the observed conditional proportion $C_g(.05)/S_g(.05)$ among significant results. Both pool one-sided and two-sided/unrestricted tests. The overall rate increases approximately linearly with nominal $\alpha \in (0, .25)$; reported values and upper bounds at .05 produce a jump, and numerical reports may be rounded. This descriptive relationship is not an estimate of the type I error rate.

At $\alpha = .05$, 493 of 6,474 NHST-significant results have applicable eJAB > 1 when calculated from reported values or bounds; these are screening flags for evidential disagreement, subject to the reporting limitations below. Although lower thresholds have been recommended in the literature (e.g., Benjamin et al., 2018; Johnson, 2013), $\alpha = .05$ remains germane in CTG: in a random sample of 52 trial protocols, 75% (SE 6%) explicitly set $\alpha \geq .05$ as the target evidential threshold. We provide the full catalog of 493 contradiction flags in the GitHub repository.

The observed conditional proportions are 6/94 (6.38%) in Early Phase I, 28/927 (3.02%) in Phase

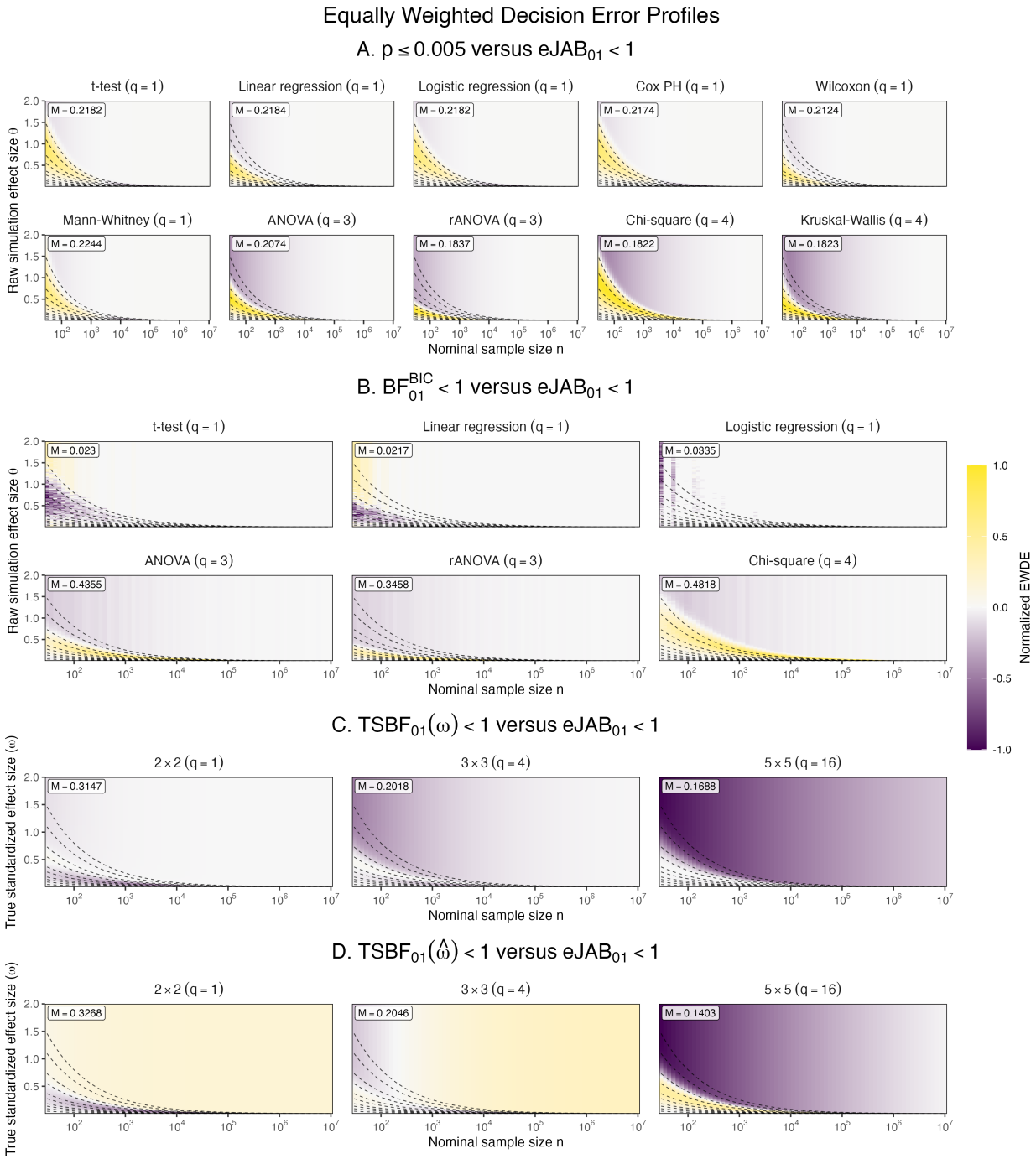


FIG 2. Equally Weighted Decision Error Profiles. (A) $p \leq .005$ versus $eJAB_{01} < 1$ across ten statistical tests. (B) $BF_{01}^{\mathrm{BIC}} < 1$ versus $eJAB_{01} < 1$ across six parametric-likelihood tests. (C) $TSBF_{01}(\omega) < 1$ versus $eJAB_{01} < 1$ and (D) $TSBF_{01}(\hat{\omega}) < 1$ versus $eJAB_{01} < 1$, with $\hat{\omega} = \sqrt{X^2/n}$, for 2×2, 3×3, and 5×5 contingency tables. Colors indicate normalized EWDE: positive values (yellow) favor eJAB, whereas negative values (purple) favor the comparison rule. Each panel reports its normalization constant M, the maximum absolute EWDE difference over its displayed grid. Dashed curves represent local alternatives of the form $\theta = c/\sqrt{n}$, with $\theta = \omega$ for the contingency-table comparisons.

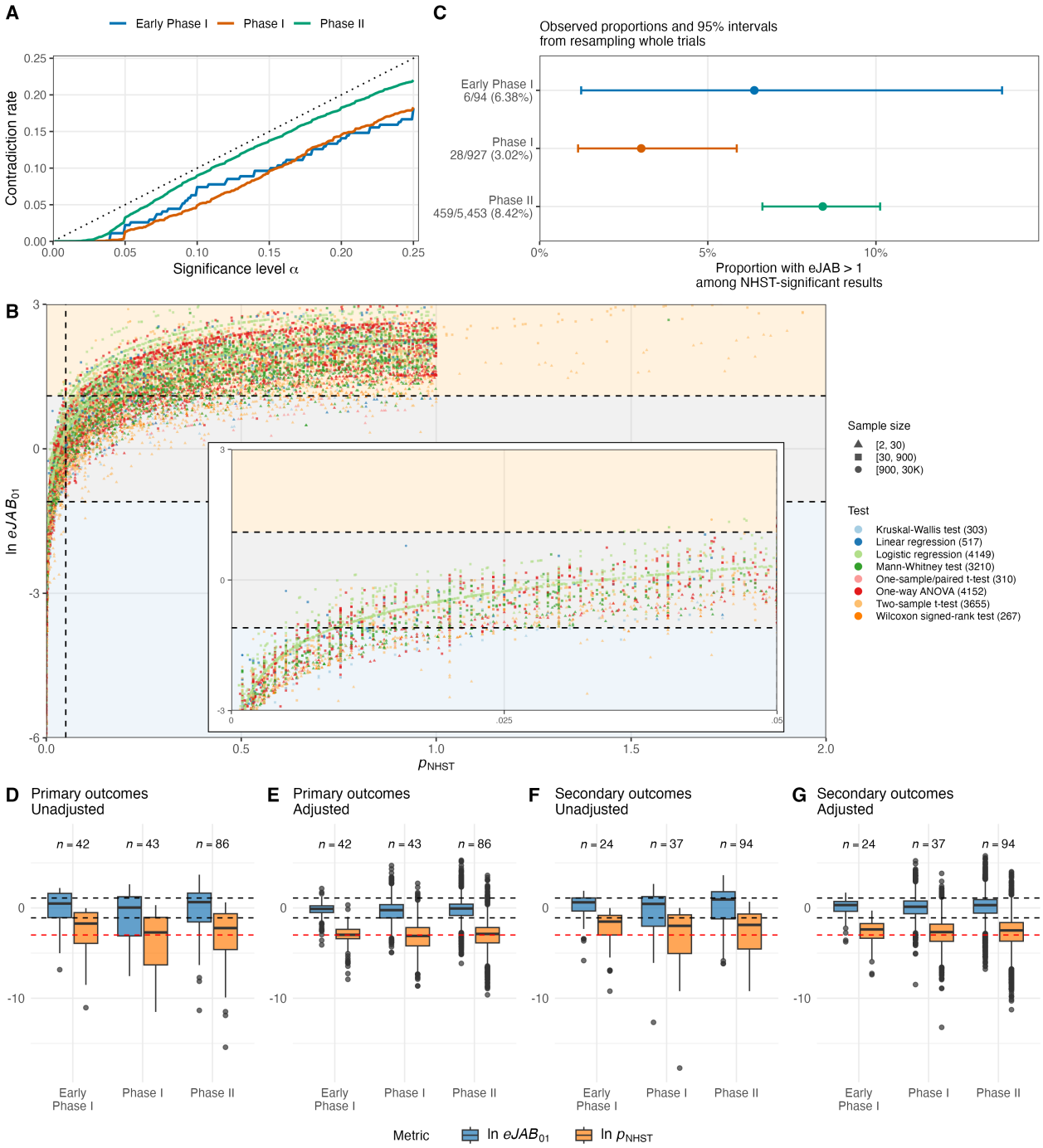


FIG 3. Clinical-trial application summaries for primary and secondary outcomes in exact Early Phase I, Phase I, and Phase II trials, combining one-sided and two-sided/unrestricted tests. For NHST comparisons, $p_{\mathrm{NHST}} = p$ for two-sided tests and $p_{\mathrm{NHST}} = 2p_1$ for one-sided tests. (A) Fraction of all eligible analyses with $p_{\mathrm{NHST}} \le \alpha$ and applicable $eJAB > 1$; the dotted line marks equality with α. (B) $\ln eJAB_{01}$ versus p_{NHST}, with colors indicating statistical tests and point shapes indicating sample-size groups. Dashed lines mark $p_{\mathrm{NHST}} = .05$ and $eJAB_{01} = 1/3$ and 3; the inset shows $p_{\mathrm{NHST}} \le .05$ and $-3 \le \ln eJAB_{01} \le 3$. (C) Observed conditional contradiction proportions $C_g(.05)/S_g(.05)$, with 95% percentile intervals from 20,000 whole-trial bootstrap resamples within each phase. All eligible analyses from a sampled trial are retained together. (D, F) Unadjusted distributions of $\ln eJAB_{01}$ (blue) and $\ln p_{\mathrm{NHST}}$ (orange) for primary and secondary outcomes, respectively. (E, G) Corresponding distributions after adjustment using linear mixed models. Panels D–G share the same y-axis scale; average analysis sample sizes appear above each phase group. Black dashed lines mark $-\ln 3$ and $\ln 3$, and red dashed lines mark $\ln 0.05$.

I, and 459/5,453 (8.42%) in Phase II, or 493/6,474 (7.62%) overall. Figure 3C shows these proportions with 95% percentile intervals of 1.23–13.75%, 1.14–5.86%, and 6.62–10.12%, respectively, from 20,000 bootstrap

resamples of whole trials within each phase. Resampling trials retains the analyses and shared comparisons within each sampled trial together. The observed proportions decrease from Early Phase I to Phase I and increase in Phase II; this descriptive pattern does not establish a phase effect. An audit of the previous study-level Dirichlet analysis found substantial prior influence; its posterior estimates and sensitivity analyses are reported separately in Supplementary Appendix C.8.

With the threshold changed to 1/3, 1,994 of the 6,474 NHST-significant results (30.80%) have applicable eJAB $> 1/3$, combining both test types. This broader category includes evidence for the alternative no stronger than anecdotal as well as evidence favoring the null. It is distinct from the eJAB > 1 contradiction category and the stronger eJAB > 3 screen.

Figure 3B displays the relationship between $\ln eJAB_{01}$ and p_{NHST}. We screen for stronger disagreements using $p_{\text{NHST}} \leq .05$ and applicable eJAB > 3, corresponding to the region at or to the left of .05 and above $\ln 3$ on this plot. For an appropriate point-null test with a numerically reported p-value, this is an illustration compatible with the Jeffreys–Lindley paradox under eJAB. The interpretation depends on the reported hypothesis, sample size, and test dimension, rather than on the two numerical cutoffs alone.

The screen returns 20 records, all documented in Supplementary Table S3 (Appendix C.7) with the finding, comparison, trial identifier, sample size, test dimension, reported p-value, and calculated eJAB. Eleven records have numerically reported p-values, all from five Phase II trials. Nine report only $p < .05$, so their eJAB values calculated at .05 are upper bounds and do not establish that the unknown values exceed 3. Four of these nine arise in an equivalence or bioequivalence context, for which significance does not test an ordinary point null of no difference; they are not interpreted as point-null paradox examples. Thus the 20 records form an auditable screening catalogue, not 20 confirmed paradox instances. Numerical reports also remain subject to rounding. Additional results stratified by MeSH category appear in Appendix B (Figures S4–S6).

The approximately linear relationship describes how the frequency of Bayes/NHST contradictions changes with the nominal significance level; it does not by itself establish which findings are type I errors. To elucidate how such disagreements arise, consider a two-sided Wald test with $q = 1$ and independent and identically distributed data for fixed $n < \infty$ and $\alpha > 1 - F_{\chi^2_1}\left(\frac{n \ln n}{n-1}\right)$. A Bayes/NHST contradiction occurs when the observed effect size falls within the interval,

$$\sqrt{\frac{1}{n} Q_{\chi^2_1}(1-\alpha)} \cdot I_1^{-\frac{1}{2}}(\hat{\theta}) \leq |\hat{\theta} - \theta_0| < \sqrt{\frac{\ln n}{n-1}} \cdot I_1^{-\frac{1}{2}}(\hat{\theta}),$$

where I_1 is the Fisher information for one observation. In this regime, the observed effect size is large enough to achieve standard statistical significance at sample size n. It is also small enough for $eJAB_{01}$ to favor the null. To favor the alternative, the latter requires the observed effect size to exceed a bound of order $\mathcal{O}_p\left(\sqrt{\frac{\ln n}{n}}\right)$.

Five numerical-report examples illustrate the stronger screen; all are Phase II analyses with $q = 1$. In NCT00565812, the registered logistic-regression analysis of joint-space narrowing progression reports $p = .047$ with $n = 848$ and $eJAB_{01} = 4.06$; a separate analysis of decreased total analgesic use at month 24 reports $p = .032$ with $n = 971$ and $eJAB_{01} = 3.13$. In NCT01225822, the logistic-regression comparison of major bleeding events between BIBR 1048, 50 mg twice daily, and enoxaparin, 40 mg once daily, reports $p = .0472$ with $n = 781$ and $eJAB_{01} = 3.91$. In NCT05013008, a two-sided t-test of change in the plasma CCN family member 4 biomarker (O95388) reports $p = .044$ with $n = 921$ and $eJAB_{01} = 4.00$. In NCT00405821, a two-sided t-test comparing annual rates of change in HIV-1 viral load between arms reports $p = .05$ with $n = 440$ and $eJAB_{01} = 3.09$; equality satisfies the NHST rule used here. These five analyses come from four trials and are not independent replications. The values describe the registered comparisons and do not establish treatment efficacy or the truth of their null hypotheses.

Three additional ClinicalTrials.gov examples illustrate the broader eJAB > 1 contradiction category across trial phases, rather than the stronger eJAB > 3 screen. In Early Phase I trial NCT00494806, the reported values for first flatus were 3.16 (standard deviation [SD] .86; $n = 34$) with rocking and 3.88 (SD .80; $n = 32$) with standard care. The t-test used a total sample size of $n = 66$ and was reported as $p < .05$, with $eJAB_{01} = 1.23$ calculated at the reported upper bound; the unreported exact p-value could yield a smaller eJAB. The reported mean difference (MD) was .7279 (95% confidence interval [CI] .3174, 1.1383). In Phase I trial NCT03907683, activity after a weekday move-more message was .16 (SD .69; $n = 80$). The reported t-test had $p = .037$, with $eJAB_{01} = 1.04$. The reported MD was .1641 (95% CI .0098, .3185). In Phase II trial NCT00539539, the cardiopulmonary resuscitation (CPR) fraction was 66 (standard error [SE] 1; $n = 815$) with feedback on and 64 (SE 1; $n = 771$) with feedback off. The linear regression analysis used a total sample size of $n = 1{,}586$ and reported $p = .016$, with $eJAB_{01} = 2.19$. The reported adjusted MD was 1.9 (95% CI .4, 3.4).

We examine the distributions of $\ln eJAB_{01}$ and $\ln p_{\text{NHST}}$ by trial phase for primary and secondary outcomes in Figure 3D and F. For primary outcomes, there

is no discernible trend toward stronger evidence across early phase I, phase I, and phase II. For secondary outcomes, we again observe no phase-wise increase in evidence. Plotting p_{NHST} and sample size by phase shows that changes in p_{NHST} track changes in sample size (Appendix B, Figure S8).

To better visualize trends in evidence across phases, we fit a linear mixed model with a fixed intercept and trial-specific random effects to each of $\ln eJAB_{01}$ and $\ln p_{\text{NHST}}$, and then examine residuals re-centered by the estimated intercept (Appendix B, Figure S7). The adjusted measures are essentially uniform across phases in Figure 3E and G. For primary outcomes, most adjusted p_{NHST} values lie near .05 and above .01. Using $eJAB$, this interpretation corresponds to 'anecdotal evidence' or 'evidence that is barely worth mentioning' for both primary and secondary outcomes (Lee and Wagenmakers, 2014, p. 105). Comparing primary with secondary outcomes, we note a slight and phase-invariant shift toward greater evidence for the alternative in primary outcomes.

5. CONCLUSIONS

$eJAB$ can be interpreted as an approximate objective Bayes factor when p-values are obtained from Wald or likelihood-ratio tests. The same interpretation should hold for tests whose statistics differ from the Wald statistic by $o_p(1)$. Simulations suggest that this interpretation holds more broadly. In addition, $eJAB$ remains model-selection consistent for p-values from any test satisfying **R1** and **R2** of Theorem 2.3.

We compared the finite-sample performance of $eJAB$ with Bayes factors computed from MCMC and the Savage–Dickey density ratio for the t-test, linear regression, logistic regression, ANOVA, rANOVA, and parametric survival analysis. Under the same generalization of the unit-information prior, $eJAB$ closely approximates the MCMC-based Bayes factor. For a chi-squared test under multinomial or product-multinomial sampling, it agrees with the test-statistic Bayes factor under the alternative, unlike the latter, also registers evidence for the null when true. For nonparametric tests, namely Wilcoxon signed-rank, Mann–Whitney, and Kruskal–Wallis, $eJAB$ performs comparably to Bayes factors based on nonparametric test statistics. Our simulations also show improved performance compared to the BIC approximation to the Bayes factor when $q > 1$.

The Equally Weighted Decision Error Profiles compare $eJAB$ with true- and estimated-effect TSBFs, BIC approximations, and NHST significance rules. Relative performance varies with sample size, effect size, and dimension; the comparisons do not establish uniform decision-error dominance.

Our exploratory reanalysis retains 493 Bayes/NHST contradiction flags among 6,474 significant results from 16,563 eligible analyses in 1,790 early-phase trials. The phase-specific summaries use observed proportions and whole-trial bootstrap intervals, with prior-sensitive posterior calculations retained as an audit in the supplement. Screening for the stronger disagreement eJAB > 3 yields 11 numerical-report records and nine records based on censored p-values; the latter, including four equivalence-context records, do not establish point-null paradox instances. The documented examples illustrate how statistical significance and Bayesian evidence can disagree under a specified approximation. They motivate scrutiny of individual hypotheses and reported analyses rather than identifying type I errors or explaining later clinical-trial failures.

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SUPPLEMENTARY MATERIAL

Supplementary material and GitHub repository

Appendices A–F are provided in the Supplementary Material, including the 20-record Jeffreys–Lindley screening catalogue (Table S3, Appendix C.7) and the posterior reproducibility and prior-sensitivity audit (Appendix C.8). Additional materials, referred to as the Online Appendices, including proofs, preprocessing code, extensive simulations—including nonparametric tests—CSV files containing Bayes/NHST contradiction

flags, and code for reproducing the figures in this paper are available in the GitHub repository at https://github.com/nathoogroup/ejabclinicaltrials. An *eJAB* calculator is available at https://flxp0x-puneet-velidi.shinyapps.io/bayesfactor/.

REFERENCES

Bartoš, F. and Wagenmakers, E.-J. (2023). A general approximation to nested Bayes factors with informed priors. *Stat* **12** e600. https://doi.org/10.1002/sta4.600

Benjamin, D. J., Berger, J. O., Johannesson, M., Nosek, B. A., Wagenmakers, E.-J., Berk, R., Bollen, K. A., Brembs, B., Brown, L., Camerer, C. et al. (2018). Redefine statistical significance. *Nature Human Behaviour* **2** 6–10. https://doi.org/10.1038/s41562-017-0189-z

Berger, J. O. and Sellke, T. (1987). Testing a point null hypothesis: The irreconcilability of p values and evidence. *Journal of the American Statistical Association* **82** 112–122. https://doi.org/10.1080/01621459.1987.10478397

Bickel, D. R. (2025). A small-sample Bayesian information criterion that does not overstate the evidence, with an application to calibrating p-values from likelihood-ratio tests. *Statistical Papers* **66** 1–17. https://doi.org/10.1007/s00362-025-01682-1

Daouia, A., Stupfler, G. and Usseglio-Carleve, A. (2025). Corrected inference about the extreme Expected Shortfall in the general Max-Domain of Attraction. *Information and Inference: A Journal of the IMA* **14** iaaf023. https://doi.org/10.1093/imaiai/iaaf023

Datta, S., Guha, R., Shudde, R. and Johnson, V. E. (2025a). Bayes factors functions based on test statistics and non-local moment prior densities. *Statistics & Probability Letters* **219** 110330. https://doi.org/10.1016/j.spl.2024.110330

Datta, S., Guha, R., Shudde, R. and Johnson, V. E. (2025b). On Bayes factor functions. *Bayesian Analysis* **20** 1399–1427. https://doi.org/10.1214/25-BA1562

Evans, M. (2016). Measuring statistical evidence using relative belief. *Computational and Structural Biotechnology Journal* **14** 91–96. https://doi.org/10.1016/j.csbj.2015.12.001

Grünwald, P. D. (2024). Beyond Neyman–Pearson: E-values enable hypothesis testing with a data-driven alpha. *Proceedings of the National Academy of Sciences* **121** e2302098121. https://doi.org/10.1073/pnas.2302098121

Halsey, L. G. (2019). The reign of the p-value is over: What alternative analyses could we employ to fill the power vacuum? *Biology Letters* **15** 1–8. https://doi.org/10.1098/rsbl.2019.0174

Held, L. and Ott, M. (2016). How the maximal evidence of p-values against point null hypotheses depends on sample size. *The American Statistician* **70** 335–341. https://doi.org/10.1080/00031305.2016.1209128

Held, L. and Ott, M. (2018). On p-values and Bayes factors. *Annual Review of Statistics and Its Application* **5** 393–419. https://doi.org/10.1146/annurev-statistics-031017-100307

Huisman, L. (2023). Are P-values and Bayes factors valid measures of evidential strength? *Psychonomic Bulletin & Review* **30** 932–941. https://doi.org/10.3758/s13423-022-02205-x

Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLOS Medicine* **2** 696–701. https://doi.org/10.1371/journal.pmed.1004085

JASP (2026). JASP (Version 0.98.0)[Computer software].

Jeffreys, H. (1936). Further significance tests. *Mathematical Proceedings of the Cambridge Philosophical Society* **32** 416–445. https://doi.org/10.1017/S0305004100019125

Johnson, V. E. (2005). Bayes factors based on test statistics. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **67** 689–701. https://doi.org/10.1111/j.1467-9868.2005.00521.x

Johnson, V. E. (2008). Properties of Bayes factors based on test statistics. *Scandinavian Journal of Statistics* **35** 354–368. https://doi.org/10.1111/j.1467-9469.2007.00576.x

Johnson, V. E. (2013). Revised standards for statistical evidence. *Proceedings of the National Academy of Sciences* **110** 19313–19317. https://doi.org/10.1073/pnas.1313476110

Johnson, V. E., Pramanik, S. and Shudde, R. (2023). Bayes factor functions for reporting outcomes of hypothesis tests. *Proceedings of the National Academy of Sciences* **120** e2217331120. https://doi.org/10.1073/pnas.2217331120

Kass, R. E. and Raftery, A. E. (1995). Bayes factors. *Journal of the American Statistical Association* **90** 773–795. https://doi.org/10.1080/01621459.1995.10476572

Kass, R. E. and Wasserman, L. (1995). A reference Bayesian test for nested hypotheses and its relationship to the Schwarz criterion. *Journal of the American Statistical Association* **90** 928–934. https://doi.org/10.2307/2291327

Kooperberg, C. and Stone, C. J. (1992). Logspline density estimation for censored data. *Journal of Computational and Graphical Statistics* **1** 301–328. https://doi.org/10.1080/10618600.1992.10474588

Lee, M. D. and Wagenmakers, E.-J. (2014). *Bayesian cognitive modeling: A practical course*. Cambridge University Press. https://doi.org/10.1017/CBO9781139087759

Ly, A., Verhagen, J. and Wagenmakers, E.-J. (2016). An evaluation of alternative methods for testing hypotheses, from the perspective of Harold Jeffreys. *Journal of Mathematical Psychology* **72** 43–55. https://doi.org/10.1016/j.jmp.2016.01.003

Marsman, M. and Wagenmakers, E.-J. (2017). Three insights from a Bayesian interpretation of the one-sided P value. *Educational and Psychological Measurement* **77** 529–539. https://doi.org/10.1177/0013164416669201

Masson, M. E. J. (2011). A tutorial on a practical Bayesian alternative to null-hypothesis significance testing. *Behavior Research Methods* **43** 679–690. https://doi.org/10.3758/s13428-010-0049-5

Miron, L., Gonçalves, R. S. and Musen, M. A. (2020). Obstacles to the reuse of study metadata in ClinicalTrials.gov. *Scientific Data* **7** 1–14. https://doi.org/10.1038/s41597-020-00780-z

Morey, R. D., Romeijn, J.-W. and Rouder, J. N. (2016). The philosophy of Bayes factors and the quantification of statistical evidence. *Journal of Mathematical Psychology* **72** 6–18. https://doi.org/10.1016/j.jmp.2015.11.001

Morey, R. D. and Rouder, J. N. (2026). BayesFactor: Computation of Bayes factors for common designs. R package version 0.9.12-4.8.

Mulder, J., Wagenmakers, E.-J. and Marsman, M. (2022). A generalization of the Savage–Dickey density ratio for testing equality and order constrained hypotheses. *The American Statistician* **76** 102–109. https://doi.org/10.1080/00031305.2020.1799861

Nathoo, F. S. and Masson, M. E. J. (2016). Bayesian alternatives to null-hypothesis significance testing for repeated-measures designs. *Journal of Mathematical Psychology* **72** 144–157. https://doi.org/10.1016/j.jmp.2015.03.003

Neath, A. A. and Cavanaugh, J. E. (2012). The Bayesian information criterion: Background, derivation, and applications. *WIREs Computational Statistics* **4** 199–203. https://doi.org/10.1002/wics.199

Raftery, A. E. (1995). Bayesian model selection in social research. *Sociological Methodology* **25** 111–163. https://doi.org/10.2307/271063

Rostgaard, K. (2023). Simple nested Bayesian hypothesis testing for meta-analysis, Cox, Poisson and logistic regression models. *Scientific Reports* **13** 1–11. https://doi.org/10.1038/s41598-023-31838-8

Sinharay, S. and Stern, H. S. (2005). An empirical comparison of methods for computing Bayes factors in generalized linear mixed models. *Journal of Computational and Graphical Statistics* **14** 415–435. https://doi.org/10.1198/106186005X47471

Stan (2025). RStan: the R interface to Stan. R package version 2.32.7.

Volinsky, C. T. and Raftery, A. E. (2000). Bayesian information criterion for censored survival models. *Biometrics* **56** 256–262. https://doi.org/10.1111/j.0006-341x.2000.00256.x

Wagenmakers, E.-J. (2007). A practical solution to the pervasive problems of p values. *Psychonomic Bulletin & Review* **14** 779–804. https://doi.org/10.3758/bf03194105

Wagenmakers, E.-J. (2022). Approximate objective Bayes factors from p-values and sample size: The $3p\sqrt{n}$ rule. *PsyArXiv* **1** 1–50. https://doi.org/10.31234/osf.io/egydq

Wagenmakers, E.-J., Bartoš, F. and Aust, F. (2026). Approximate Bayes factors are approximate e-processes. *SSRN*. https://doi.org/10.2139/ssrn.6138254

Wagenmakers, E.-J. and Ly, A. (2023). History and nature of the Jeffreys–Lindley paradox. *Archive for History of Exact Sciences* **77** 25–72. https://doi.org/10.1007/s00407-022-00298-3

Wasserstein, R. L. and Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician* **70** 129–133. https://dx.doi.org/10.1080/00031305.2016.1154108

Wei, Z., Nathoo, F. S. and Masson, M. E. J. (2023). Investigating the relationship between the Bayes factor and the separation of credible intervals. *Psychonomic Bulletin & Review* **30** 1759–1781. https://doi.org/10.3758/s13423-023-02295-1

Yuan, Y. and Johnson, V. E. (2008). Bayesian hypothesis tests using nonparametric statistics. *Statistica Sinica* **18** 1185–1200.